

# BIO504P PRACTICAL

## FINAL TERM-2024- 100%

### BIOCHEMISTRY I

BY SULMAN ALI GUJJAR

0341-7547044

## OBJECTIVES

- 1- Monosaccharides react **\*more\*** whereas disaccharides and Polysaccharides react slowly.
- 2- An amino group and a carboxyl group bonded to the **\* $\alpha$  carbon\***.
- 3- Within the range, there is a temperature value at which the enzyme shows maximum activity. This point for an enzyme is called as **\*optimum temperature\***
- 4- Long chain fatty acids found in fats have low saponification value because they have a relatively **\*fewer number of carboxylic functional groups\*** per unit mass of the fat.
- 5- The rate of Movement of a biomolecule during paper chromatography is known as its **\*(relative mobility and retardation factor both)\***.
- 6- Amino acids give blue/purple color when they react with **\*ninhydrin\***.
- 7- The primary characteristic of lipid that distinguish them from other compounds is **\*presence of hydrocarbon\***.
- 8- On the hydrolysis triglycerides yields **\*glycerol and soap (fatty acid)\***.
- 9- For the preparation of 10% NaCl solution, the amount of NaCl required **\*(10 gm NaCl in 100 ml of water\*)**
- 10- The rate of movement of a biomolecule during paper chromatography is known as its **\*relative mobility (Rf)**.
- 11- Enzyme assays are used for .... **Enzyme activity**
- 12- The spot is also **translucent** in nature ?
- 13- Reagent required for iodine test. **Iodine solution: Dissolve Iodine (1.0 g) and potassium iodide (2.0 g) in 300 ml of distilled.**
- 14- How can find the value of amino acid in chromatography . **By spraying ninhydrin .**
- 15- Nature of chromatographic solvent . **\*Polar \***
- 16- In monisch test which is used for color product. **\*Ethanol\***
- 17- For detection of reducing suger test name. **\*Benedict Test\***
- 18- The process of saponification produces. **\*Soap**
- 19- Enzyme assays are the methods used in laboratory for measuring. **Enzyme Activity**
- 20- The process of soap making is called... **Saponification**
- 21- The relative amount of components of solution is called. .... **Concentration**
- 22- Benedict test is used for detection of ..... **Reducing sugar**
- 23- Enzymes essay are used to measure. **Activity**
- 24- In Molisch test, which reagent is used to give coloured products. **Ethanol.**
- 25- The degree of movement of biomolecules up the paper is associated to it's relative affinity for paper which is. .... In nature? **Hydrophilic.**
- 26- The long chain fatty acids in fats have low saponification value because they have ..... Per unit mass of fat. **Answer: few number of carboxylic functional groups.**

- 27- Within the range, there is a temperature value at which the enzyme shows maximum activity. This point for an enzyme is called as -----\***optimum temperature**\*
- 28- On the hydrolysis triglycerides yields ----- **\*glycerol and soap\***.
- 29- Monosaccharides react **\*more\*** whereas disaccharides and Polysaccharides react slowly.
- 30- -An amino group and a carboxyl group bonded to the **\*α carbon\***.
- 31- Within the range, there is a temperature value at which the enzyme shows maximum activity. This point for an enzyme is called as **\*optimum temperature\***
- 32- Long chain fatty acids found in Fats have low saponification value because they have a relatively **\*fewer number of carboxylic Functional groups\*** per unit mass of the fat.
- 33- The rate of Movement of a biomolecule during paper chromatography is known as its **\*(relative mobility and retardation factor both)\***
- 34- Amino acids give blue/purple color when they react with **\*ninhydrin\*** .
- 35- The primary characteristic of lipid that distinguish them from other compounds is **\*presence of hydrocarbon\***.
- 36- On the hydrolysis triglycerides yields **\*glycerol and soap\***.
- 37- For the preparation of 10% NaCl solution, the amount of NaCl required **\*(10 gm NaCl in 100 ml of water\*)**
- 38- Nature of chromatographic solvent ? **\*Hydrophobic \***
- 39- In Molisch test which is used for colour product?... ethanol
- 40- Result of saponification? **Bubble formation**
- 41- For detection of reducing sugar test name? **Benedict**
- 42- In chromatography, amino acids give blue or purple color when they react with...  
**Ninhydrin**
- 43- The process of saponification produces... **soap**
- 44- Enzyme assays are the methods used in laboratory for measuring.... **Soap**
- 45- The process of soap making is called...**Saponification**
- 46- Which give color in molisch test? **Ethanol**
- 47- Triglycerides produce(soap and glycerol
- 48- Low saponification(low carboxylic acid)
- 49- Cellulose and glycogens are.... **polysaccharides**
- 50- solute per liter Molarity: **molarity**
- 51- Purple colour formed in protein.. **ninhydrin**
- 52- The relative amount of components of solution is called..... **Concentration**
- 53- Benedict test is used for detection of ..... **Reducing sugar**
- 54- In amino acids and ..carboxylic acid is bounded by ... **amino group**
- 55- Enzymes assay are used to measure... **Activity**
- 56- On hydroxyl triacylglycerol yield.... **Glycerol and Soap**
- 57- Number of moles of solute dissolve per liter of solvent-----**molarity**
- 58- Cellulose and glycogen----polysaccharide
- 59- Movement of biomolecule during chromatography----**relative mobility**
- 60- Characteristics of lipid that distinguishes them----presence of **hydrocarbon**
- 61- Triglyceride yield----**glycerol and soap**
- 62- Volume percentage formula. **Volume Percentage of a liquid = Volume of the liquid/ Total volume x 100**
- 63- Tertiary structure is generally stabilized by **nonlocal interactions**, most commonly the formation of a hydrophobic core.
- 64- Saponification of long chain fatty acids shows **less number of carboxylic** functional groups.
- 65- **Capillary** action in chromatography.
- 66- Triacylglycerol **yield soap and glycerol** on hydrolysis.
- 67- Monosaccharides react more in **molisch test**.
- 68- Range of temperature called **optimum temperature**.
- 69- Process of hydrolysis of fats is **saponification**.
- 70- Number is “the number of milligrams of potassium hydroxide required to neutralize the fatty Acids resulting from the complete hydrolysis of 1g of fat. **Saponification number**
- 71- Hydrolysis of triglyceride’s yield.,. **soap and glycerol**
- 72- ....is the number of equivalents dissolved per liter of solution. **Normality**
- 73- Result of saponification? **Bubble formation**

- 74- For detection of reducing sugar test name? **Benedict Test.**
- 75- The degree of movement of biomolecules up the paper is associated to its relative affinity for paper which is..... In nature. **Hydrophilic.**
- 76- Phenylalanine carries a..... benzene ring. **Inactivated.**
- 77- Volume percentage of a liquid is.....  **$\text{volume of liquid/total volume} \times 100$**
- 78- The test used only for protein detection.... **Buuret test.**
- 79- An amino group and a carboxyl group bonded to the? **same carbon atom**
- 80- Within the range, there is a temperature value at which the enzyme shows maximum activity. This point for an enzyme is called as **optimal temperature**
- 81- long chain fatty acids found in fats have low saponification value because they have a relatively **\*fewer number of carboxylic functional groups\*** per unit mass of the fat.
- 82- Amino acids give blue/purple color when they react with? **ninhydrin**
- 83- Cellulose and glycogen are ...? **polysaccharides**
- 84- Low saponification value is directly proportional to ....? **Less carboxylic acid functional group**
- 85- ..... is the solution which can dissolve no more amount of the solute, at a given .....temperature. **\*Saturated solutions\***
- 86- Hydrolysis of Triacylglycerol produce.... **\*Glycerol and Soap\***
- 87- Plot a graph between temperature and Absorbance by taking temperature along ..... And absorbance along.... **\*x-axis and y-axis\***
- 88- If Biuret's reagent is added to a solution with just single peptides then.... **\*No colour change\***
- 89- Benedict's test reagents.. **\*sodium citrate, copper sulfate and sodium carbonate\***
- 90- The rate of Movement of a biomolecule during paper chromatography is known as its.... **\*relative mobility or retardation factor.\***
- 91- The degree of movement of Biomolecule up the paper is associated to its relative affinity of the solvent... **\*Hydrophobic\***
- 92- The hydrolysis of fatty acids to produce soap... **\*Saponification\***
- 93- • If pH of solution is 0, it is **\*highly acidic\***
- 94- • Cellulose and glycogen are **\*polysaccharides\***
- 95- • Test used to detect only the presence of proteins is **\*Ninhydrin test\***
- 96- • If we plot a graph between temperature and absorbance, temperature and absorbance are taken along **\*x-axis, y-axis\***
- 97- • Low saponification value is directly proportional to **\*Less carboxylic acid functional group\*** per unit mass of fat.
- 98- • On hydrolysis, triglycerides give • Glycerol and acid • **Glycerol and Soap** • water
- 99- • In chromatography, solvent move up the paper due to **\*capillary action\***
- 100- • Rate of movement of biomolecule during paper chromatography is called **\*both relative mobility and retardation factor\***
- 101- • Lipids are distinguished from on the basis of their **\*hydrophobic nature\***

## SUBJECTIVE

### Define amylase

Amylase is an enzyme that catalyzes the hydrolysis of starch into simpler carbohydrates such as maltose. It is present naturally in human saliva. Amylase activity can be determined by the use of iodine solution. Iodine reacts with starch to form a blue black complex.

### How do you know about lipids? Explain briefly. (3 marks).

The **lipids** are a heterogeneous group of compounds that contain hydrocarbons and include fats, oils, steroids, waxes, their derivatives and precursors. They are insoluble in water but are soluble in nonpolar solvents such as ether and chloroform.

**Iodine test** This test is used to differentiate polysaccharides from monosaccharides and disaccharides.

**Principle** Different carbohydrates e.g. starch form colored adsorption complexes with iodine due to the adsorption of iodine on the polysaccharide chains. The intensity of the color is dependent on the length of the unbranched or linear chain accessible for the complex formation. Iodine forms complex with starch to give deep blue-black color. Glycogen (highly branched polysaccharide) gives red color with iodine. **Reagents Required** 1. Iodine solution: Dissolve Iodine (1.0 g) and potassium iodide (2.0 g) in 300 ml of distilled water. 2. Starch solution (1%) **Procedure** 1. Add 2 ml of sample in a test tube. 2. Add 1-2 drops of iodine solution. 3. Formation of blue black color indicates the presence of starch

**Xanthoproteic Test** It is a general test used for the detection of tyrosine, tryptophan and proteins containing these two amino acids. **Reagents Required** i. Conc. Nitric acid ii. 40% sodium hydroxide solution iii. Amino acid solution (Tyrosine/Tryptophan)

**An amino group and a carboxyl group bonded to the?**

**Spot Test Procedure** i. Take a piece of filter paper ii. Add a drop of sample on it iii. A greasy spot penetrating the paper indicates lipids. The spot is also translucent in nature.

**Saponification test** Saponification is the hydrolysis of fats or oils under basic conditions to give glycerol and the salt of the corresponding fatty acid. **Principle** Saponification is the process that produces soap from fats. In this process, oil or fat is hydrolyzed with alkali that results in the formation of glycerol and salt of fatty acids (soap). **Procedure** i. Take sample and alcoholic potassium hydroxide in 1:3 ratios. ii. Boil for 10-20 minutes. iii. After heating, a semi solid layer develops (soap). iv. Separate the upper semi solid layer. v. Add little water to upper separated layer and shake well. Formation of bubbles indicates soap formation.

**Define chemical solutions and name their types?**

**Iodine solution and dilute iodine solution**

1. **Enzyme solution** Add one ml of saliva in a beaker and dilute it by adding 9 ml of water.
2. **Starch solution (2%)** Prepare stock starch solution by dissolving 2 g of starch in 70 ml of distilled water by heating on hot plate and then raise the volume to 100 ml with distilled water
3. **Stock Iodine Solution** Add Iodine (5 g) and KI (50 g) to 500 ml of distilled water and raise the volume to 1 Litre with distilled water.
4. **Dilute Iodine Solution** Take 1 ml stock iodine solution and raise the volume to 100 ml with distilled water.

**Define chemical reaction and write their types? 3 marks**

**Saturated, unsaturated and supersaturated solution**

**Solution** A solution is a homogenous mixture of atoms, ions or molecules of two or more substances. A homogenous mixture is that which has uniform composition throughout its body. Types of solution **Unsaturated solution** A solution that is capable of dissolving more solute at a given temperature than it already contains, is known as unsaturated solution. **Saturated solution** A saturated solution is the solution which can dissolve no more amount of the solute, at a given temperature. **Supersaturated solution** A solution that contains more dissolved solute than a saturated solution is called super saturated solution

**Define chromatography and chromatogram/ Movement of biomolecules and organic solvents in chromatography/ What is retardation factor and how is it calculated? - How amino acids are made visible on chromatogram?**

**Chromatography** is an analytical method used for separation of different biomolecule on the basis of their chemical properties. In paper chromatography, a biomolecule (or mixture of biomolecules) is spotted on a piece of filter paper and is placed in an organic solvent. The hydrophobic organic solvent moves up the paper by capillary action. As the solvent reaches the biomolecule, the biomolecule starts to move up the paper. The degree of movement of biomolecule up the paper is associated to its relative affinity for paper (hydrophilic in nature) and the solvent (hydrophobic in nature). Paper chromatography is very valuable method for characterizing amino acids. Different amino acids migrate at different rates on the paper because

of difference in their R groups. The rate of movement of a biomolecule during paper chromatography is known as its relative mobility (Rf). Rf is also known as **retardation factor**. Rf is simply the distance the biomolecule moved through the filter paper divided by the distance the moved by solvent through the paper.  $Rf = \text{Distance travelled by sample} / \text{Distance travelled by solvent}$

**Ninhydrin Test** It is a general test used for the detection of proteins, peptides and amino acids.

**Principle** Ninhydrin reacts with  $\alpha$ -amino acids ( $-\text{NH}_2$ ) to give a purple colored complex.

Proteins also have a free amino group on the  $\alpha$  carbon that can react with ninhydrin to give a blue purple product. Amino acids such as Proline which has a secondary amino group react with ninhydrin to form a yellow reaction product. **Reagents Required** 1. Ninhydrin solution: Add 0.2 g of ninhydrin in 50 ml of acetone and raise the volume to 100 ml with acetone. 2. Amino acid solution (Lysine/alanine/proline etc.)

**Procedure** 1. Add 1 ml of amino acid solution in a test tube. 2. Add 5 drops of 0.2% ninhydrin solution. 3. Place the test tubes in boiling water for 5 min. 4. Allow it to cool and observe the purple color formed.

**Biuret Test** This test is used for the detection of proteins containing at least two peptide bonds.

**Principle** The reaction incorporates the violet complex formation of the proteins with  $\text{Cu}^{2+}$  ions in an intensely alkaline solution. **Reagents Required** 1. 10% sodium hydroxide solution 2. 1% copper sulfate solution 3. Protein solution (casein, gelatin, albumin)

**Procedure** 1. Add 1 ml of protein solution in a test tube. 2. Add 1 ml of 10% sodium hydroxide solution and 2-3 drops of 1% copper sulfate solution. 3. Mix well and observe the formation of violet color.

**Molisch's test** It is a sensitive group test that is used regularly for the detection for all carbohydrates present either in free or combined form. **Principle:** This test makes use of concentrated  $\text{H}_2\text{SO}_4$  to catalyze the dehydration of sugars to form hydroxymethyl furfural (from hexoses) or furfural (from pentoses). The furfurals then condense with  $\alpha$ -naphthol to give a violet or purple colored product.

**Reagents Required** i. Conc. sulfuric acid ii. Molisch's Reagent: 5%  $\alpha$ -naphthol solution in 95% ethanol iii. Carbohydrates (glucose, maltose) **Procedure** 1. Add 1-2 ml of unknown solution in a test tube. 2. Add 2-3 drops of Molisch's reagent and mix well. 3. Hold the test tube in an inclined position and add carefully 1-2 ml of Conc. Sulfuric acid down the side of the tube to make a layer of it at the bottom of the tube. 4. Formation of purple product/ring at the border of two layers indicates a positive test.

**Proteins** are organic compounds made of amino acids. The amino acids joined together by the peptide bonds. Proteins have three dimensional structures. **Primary structure** is the amino acid sequence. **Secondary structure**, alpha helix, beta sheet. Secondary structures are local many regions of different secondary structure can be present in the same protein molecule. It regularly repeats local structures stabilized by hydrogen bonds. **Tertiary structure** is generally stabilized by nonlocal interactions, most commonly the formation of a hydrophobic core, but also through salt bridges, hydrogen bonds, disulfide bonds, and even post-translational modifications.

**Quaternary structure**, formed by several protein Molecules usually called protein subunits

**How 5% NaCl solution prepare write it's procedure? 5 marks**

**Procedure i. Preparation of 5% NaCl solution** 1. Weigh 5 grams of sodium chloride. 2.

Dissolve the sodium chloride ( $\text{NaCl}$ ) in 25 ml of distilled water in a beaker with the help of stirrer. 3. Transfer the contents into a volumetric flask. Add some more water (10-20 ml) to the beaker and rinse the walls of beaker and add that to the same flask. 4. Raise the volume to 100 ml in the volumetric flask to prepare 5% NaCl solution. 5. Check the pH of the solution with pH meter.

**Carbohydrates** are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis. **Monosaccharides** (simple sugars) are those which cannot be hydrolyzed further into simpler forms. The backbones of monosaccharides are unbranched carbon chains in which all the carbon atoms are connected by single bonds, for example glucose, fructose.

**Disaccharides** are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis. For example, maltose, lactose. **Oligosaccharides** are small chains of monosaccharide units (3-10) linked by distinctive linkages called glycosidic linkage.

**Polysaccharides** (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis. Some have hundreds or thousands of units. For example cellulose, glycogen

### #Prepare 10% HCl solution (100ml) from 37% HCl ?

1. Take 50 ml of distilled water in a volumetric flask. 2. Calculate the volume of HCl required by using the formula.  $C_1V_1=C_2V_2$   $37 \times V_1=10 \times 100$   $V_1= 1000/37 = 27.02$  ml 3. Add 27.02 ml of HCl to the volumetric flask slowly and raise the volume to 100 ml with distilled water to prepare 100 ml of 10% HCl solution. 4. Record the pH of the solution using pH meter

**Benedict's test** is used for the detection of reducing sugars (sugars having free reactive carbonyl group). **Principle** This test utilizes a mixture of sodium citrate, copper (II) sulfate and sodium carbonate in a slightly basic solution. Reducing sugars reduce the copper (II) ions to copper (I) oxide (Red ppt).  $R-CHO + 2Cu^{2+} + 5OH^- \rightarrow R-CO_2^- + Cu_2O \text{ (red ppt)} + 3H_2O$   $R-CHO$  =Reducing carbohydrates  $R-CO_2^-$  = Carbohydrate ion **Reagents Required** i. Benedict's Reagent: Dissolve Sodium Citrate (173 g) and anhydrous Sodium Carbonate (100 g) in about 700 ml of distilled water by gently heating the contents. Dissolve Copper Sulfate (17.3 g) in about 100 mL of distilled water in a separate beaker. Transfer this solution gradually into the Carbonate-Citrate mixture with constant stirring and raise the volume to 1 L with distilled water. ii. Carbohydrates (glucose, starch/sucrose) **Procedure** 1. Add 1 ml of sample in a test tube. 2. Then add 2 ml of Benedict's reagent. 3. Heat the solution in a boiling water bath for 3 minutes. 4. Formation of red precipitate indicates positive test.

#### • What is enzyme activity. List factors affecting enzyme activity.

Enzyme activity is a measure of quantity of active enzyme present. Different factors that affect enzyme activity include pH, temperature and substrate concentration.

#### Difference between solute, solvent

The solvent is the component of a solution that is visualized as dissolving another component called a solute. Usually the component present in the larger quantity is called the **solvent**, and the component present in the smaller quantity is called the **solute**.

**Molarity (M)** Molarity or the molar concentration is the number of moles of solute dissolved per dm<sup>3</sup> of solution. (1 dm<sup>3</sup> is equal to 1 Liter)  $\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in liters}}$  **Molality (m)** Molality is the number of moles of solute dissolved per kilogram of solvent.  $\text{Molality} = \frac{\text{Number of moles of solute}}{\text{Kg of solvent}}$  **Normality (N)** Normality is the number of equivalents dissolved per liter of solution.  $\text{Normality} = \frac{\text{No. of equivalents}}{\text{Volume of solution in liters}}$

#### - Effect of temperature on enzymatic activity, with procedure.

**Effect of temperature on enzyme activity. Procedure** i. Add 5 ml of 1% starch solution (pH: 7.0) in 7 test tubes and place at different temperature (30- 90oC) for equilibration. ii. Add 1 ml of amylase solution to each test tube. iii. Keep the test tubes in water bath for 10 minutes at different temperature (30-70oC with increments of 10oC). iv. Add 0.5 mL of the reaction mixture to 5 mL of the 0.1 N HCl stopping solution. v. Add 0.5 mL of the above mixture to 5 mL iodine solution to develop color. Shake and mix. vii. Measure the absorbance with a spectrophotometer at 620 nm setting the spectrophotometer to zero with distilled water. vii. Plot a graph between temperature and Absorbance by taking temperature along x-axis and absorbance along y-axis.

#### What is concentration ? Write four units

**Concentration and its units** **Concentration** means the relative amounts of the components of a solution. It tells the ratio of the quantity of one component to the quantity of the other or to the total quantity of solution. It has many units. Some common units are discussed below. **Mass Percentage, Mass of solution, Parts per Million (ppm), Molarity (M), Molality (m), Normality (N)**

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#### Q3. What do you know about amino acids?

**Amino acids** have an amino group and a carboxyl group bonded to the same carbon atom (the  $\alpha$  carbon) and a side chain that varies for different amino acids. Proteins are polymers of amino

acids, with each amino acid residue joined to its neighbor by a specific type of covalent bond termed the peptide bond.

**Analysis of protein and detection of protein**

difference between optimum pH and optimum temperature.

**An optimum temperature** is a temperature at which a catalyst is most effective. Within the range, there is a pH value at which the enzyme shows maximum activity. This point is known as **optimum pH** for enzyme